Kevin Nam

Seoul, South Korea | rallyk3vin@gmail.com | +82 10 2443 7250 | rallykevin.github.io | github.com/rallykevin

About Me

I am currently pursuing a combined MS/PhD in the Security Optimization Research Lab at Seoul National University. I hold a Bachelor of Science in Electrical and Computer Engineering from the same university, with a focus on hardware/system/AI security and privacy. But my research fields are not limitied to them. Back when I was an undergraduate student, I had been deeply passionate about hardware design and computer architecture. My first research topics were naturally related to efficient system integration (including hardware designs) for high-performance domain-specific computation. My research interests have expanded since then, including security and privacy-preserving computation. My current focused research topics are homomorphic encryption, multi-party computation, and trusted execution environments. As I deepen my expertise in these areas, I remain open to exploring new challenges and interdisciplinary fields within broader scopes.

Education

Seoul National University, BS in Electrical and Computer Engineering — Mar 2014 – Feb 2020

- Left for military service (ROKAF) during 2017-2019

Seoul National University, MS/Ph.D in Electrical and Computer Engineering — Mar 2020 – Feb 2026 (expected)

Publications

Refereed Conference Publications

[1] **[Security'25]** " SLOTHE: Lazy Approximation of Non-Arithmetic Neural Network Functions over Encrypted Data". *Kevin Nam*, Youyeon Joo*, Seungjin Ha, and Yunheung Paek. (* co-first authors) In USENIX Security Symposium (Security'25), August 2025.*
(KIISE S, BK21 IF: 3, CSRankings).

[2] **[ASPLOS'25]** "Affinity-based Optimizations for TFHE on Processing-in-DRAM". *Kevin Nam, Heonhui Jung, Hyunyoung Oh, and Yunheung Paek. In the 30th International Conference on Architectural Support for Programming Languages and Operating Systems (ASPLOS'25), April 2025.*
(KIISE S, BK21 IF: 4, CSRankings).

[3] **[Security'25]** "LOHEN: Layer-wise Optimizations for Neural Network Inferences over Encrypted Data with High Performance or Accuracy". *Kevin Nam, Youyeon Joo, Dongju Lee, Seungjin Ha, Hyunyoung Oh, Hyungon Moon, and Yunheung Paek. In USENIX Security Symposium (Security'25), August 2025.*
(KIISE S, BK21 IF: 3, CSRankings).

[4] **[ICCAD'22]** "Accelerating N-bit Operations over TFHE on Commodity CPU-FPGA". *Kevin Nam, Hyunyoung Oh, Hyungon Moon, and Yunheung Paek. In IEEE/ACM International Conference on Computer-Aided Design (ICCAD'22), November 2022.*
(KIISE A, BK21 IF: 3, CSRankings).

[5] "Area-Efficient Accelerator for the Full NTRU-KEM Algorithm". *Yongseok Lee, Kevin Nam, Youyeon Joo, Jeehwan Kim, Hyunyoung Oh, Yunheung Paek. In International Conference on Computational Science and Its Applications (ICCSA'23). Cham: Springer Nature Switzerland, 2023.*

Refereed Journal Publications

[6] "An Efficient Hardware/Software Co-design for FALCON on Low-End Embedded Systems". *Yongseok Lee, Jonghee Youn, Kevin Nam, Heon Hui Jung, Myunghyun Cho, Jimyung Na, Jong-Yeon Park, Seungsu Jeon, Bo Gyeong Kang, Hyunyoung Oh, Yunheung Paek. In IEEE Access, 2024.*

[7] "MeetGo: A trusted execution environment for remote applications on FPGA". *Hyunyoung Oh, Kevin Nam, Seongil Jeon, and Yeongpil Cho, Yunheung Paek. In IEEE Access, 2021.*

Domestic (Only First Author Papers)

[8] "Partially Homomorphic Encryption HW accelerator". *Kevin Nam, Jiwon Chang, Myunghyun Cho, Inyoung Bang, Yunheung Paek. Annual Conference of KIPS, 2020*

[9] "Realization of Homomorphic Encrypted Deep Learning Models". *Kevin Nam, Myunghyun Cho, Hyunjun Kim, Yunheung*

Paek. Annual Conference of KIPS, 2021,* **(Undang Academic Award)**

[10] "Side-Channel Attacks on Homomorphic Encryption and Their Mitigation Methods". *Kevin Nam, Youyeon Joo, Yunheung Paek. Annual Symposium of KIPS, 2023* **(NIPA Director Award)**

[11] "Quantization of Non-Arithmetics to Optimize CNNs over TFHE". *Kevin Nam, Heonhui Jung, Dongju Lee, Yunheung Paek. Annual Symposium of KIPS, 2024*

[12] "Intermediate Data Guided Approximation for Fast and Accurate Encrypted Neural Networks". *Kevin Nam, Youyeon Joo, Seungjin Ha, Yunheung Paek. Annual Conference of KIPS, 2024.*

Patents

[13] "METHOD FOR PROCESSING HOMOMORPHIC CIPHIERTEXT AND ELECTRONIC APPARATUS". *Yunheung Paek, Heonhui Jung, Kevin Nam, Seungjin Ha, Junbum Shin, Inkwan Yu, Sunchul Jung, Jungjoo Seo. Feb 2025. US Patent No. 19/052442, Applied*

[14] "METHOD FOR PROCESSING HOMOMORPHIC CIPHIERTEXT AND ELECTRONIC APPARATUS". *Yunheung Paek, Heonhui Jung, Kevin Nam, Seungjin Ha, Junbum Shin, Inkwan Yu, Sunchul Jung, Jungjoo Seo. Jan 2025. KR Patent No. 10-2025-0008424, Applied*

[15] "METHOD AND APPRATUS FOR COMPUTING ENCRYPTED DATA USING MULTI-HOMOMORPHIC ENCRYPTION SYSTEM". *Yunheung Paek, Kevin Nam, Youyeon Joo. Nov 2024. KR Patent No. 10-2024-0153707, Applied*

[16] "TEE ENVIRONMENT PROVIDING APPARATUS AND METHOD USING FPGA". *Yunheung Paek, Hyunyoung Oh, Kevin Nam, Yeongpil Cho. Jan 2021. KR Patent No. 10-2021-0006281, Applied*

Research Experience (w. fundings, PI : Yunheung Paek)

Zero-Error Privacy-Preserving Neural Networks *funded by NRF, South Korea* — Oct 2023 – present

- Research on desining efficient and accurate neural network services over encrypted data
- Integrate non-mathematical methods (e.g., compiler) with the mathematical cryptosystems
- Currently participating as the FHE project leader

Secure FHE Key Management System *funded by CryptoLab, South Korea* — Mar 2023 – April 2024

- Designed TEE-based FHE key management system
- Developed the protocol to publish and depricate keys to users specifically adapted for FHE
- Participated as the TEE software engineer

Neural Network Program to FHE Transpiler *funded by ETRI, South Korea* — Oct 2023 – Nov 2023

- Designed a transpiler that changes a tensorflow, Pytorch program to FHE programs
- Developed the FHE routines that corresponds to the kernels used in tensorflow (e.g., convolution, activations)
- **Solely participated in the whole project**

PEC Platform *funded by the National Intelligence Service, South Korea* — Mar 2022 – Nov 2023

- Designed a Privacy-Enhancing-Computation (PEC) platform that exploits the use of various privacy-preserving techniques (e.g., FHE, MPC, TEE)
- Developed a framework that transpiles a normal code into privacy-preserving one, suggesting appropriate privacy-preserving technique for each partitions
- Participated as **the project leader** (designing the dataflow of the platform/framework)

FPGA Accelerator for CKKS *funded by CryptoLab, South Korea* — Mar 2022 – Nov 2022

- Designed an FPGA-based CKKS Accelerator for its bootstrapping algorithm
- Participated as **the algorithm analyzer** - designed the overall operation mapping and pipeline

FPGA Accelerator for FHE *funded by the National Intelligence Service* — Mar 2021 – Nov 2021

- Designed an FPGA-based TFHE accelerator exploiting the double parallelism of TFHE
- Developed a methodology to find the optimal setup of parallelism considering the system's memory bandwidth
- **Solely participated in the whole project**, from algorithm analysis to system integration

Honors & Awards

Undang Academic Award *from KIPS, Best Graduate Student Paper Award* — Dec 2021

NIPA Director Award *from ASK 2023* Jun 2023

Award for Excellence in Teaching Assistant, *from Seoul National University* Jan 2024
Logic circuit Design Course

Scholarship
BK 21+ Scholarship *by the Ministry of Education of Korea* Mar 2020 - present
SNU graduate student scholarship *by Seoul National University* Mar 2022 - Feb 2024

Activities

EuroSys'25 *Shadow Program Committee* 2025

Teaching (Assistant) Experience

Seoul National University
Logic Circuit Design Course Sep 2023 - Dec 2023
Data Security & Privacy Sep 2024 - Dec 2024

Skills

Languages: English, French, Korean

Programming Languages: C/C++, Python, HDL, JAVA

Tools/Frameworks: Vitis, Vivado, Xcellium, Synopsys DC, Tensorflow, Pytorch, Docker, Git, Linux

Privacy-Preserving Tools: MS SEAL, OpenFHE, HeaaN, TFHE-rs, OpenCheetah, and much more